

Name: key

Total: _____/100 pt.

Exam I

Stat 801

Sec 250

Instructions:

- You are allowed 2 hours (120 minutes) for this exam. Please pace yourself accordingly.
 - You may use a calculator and your 1 page formula sheet on this exam. However, you are not allowed to use your cell phone AT ALL! It is a “close book” exam.
 - The standard normal table and the t-table are attached to your exam so you DON’T need to use your own tables.
 - In order to receive full credit, you must show your work and carefully justify your answers. The correct answer without any work will receive little or no credit. Where it is applicable, draw a picture!
 - If you need more room, use the backs of the pages AND INDICATE that you have done so.
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- Please write neatly.
 - GOOD LUCK!!

1. (5pt) This fall the United States will elect its 45th president. The Republican and Democratic Party presidential hopefuls have been campaigning relentlessly in an effort to become elected the next president of the United States. A political scientist working for NORC (National Opinion Research Center) is conducting a study of public opinion among registered voters on favorability of a female president. She obtains a representative sample consisting of 1254 registered voters, 58% of which believe that the country is ready for a female president. Answer each of the following questions!

(a) What is the population of interest?

All registered voters in the US.

(b) What is the sample?

The 1254 registered voters participating in the study.

(c) What is the parameter?

The percentage of all registered voters who believe that a female president is acceptable.

(d) What is the statistic?

The percentage of the 1254 registered voters who believe that a female president is acceptable.

(e) What test statistic would you use to test whether the percentage of all registered voters who believe that a female president is acceptable is less than 50%? Write down the formula. No calculations are needed!

$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

2. **TRUE OR FALSE?** (5pt) Answer the following questions by indicating whether they are true or false. Circle the correct answer!

(a) T / ☒ F When the p-value is larger than α , we accept the null hypothesis.

(b) ☒ T / F If the population mean is known there is no need to test hypothesis concerning the population mean.

(c) T / ☒ F The center of a non-normal distribution is best described with the mean.

(d) T / ☒ F In a hypothesis test we want to maximize the power because our goal is to show that there is no significant difference between the nature and our claim assuming that there is no significant difference between the nature and the claim.

(e) ☒ T / F If we reject a null hypothesis at the 1% level, we also reject it at the 5% level.

3. The selling price of a particular brand of a high performance workstation among dealers nationwide is assumed to be normally distributed with mean $\mu = \$2,400$ and standard deviation $\sigma = \$500$.

(a) (1pt) Define the Normal random variable X.

$$X = \text{price of workstation} \sim N(2400, 500^2)$$

(b) (4pt) What is the probability that a high performance workstation of the same brand, chosen randomly from a dealer, will cost less than \$2,300?

$$\begin{aligned} P(X < 2300) &= P\left(Z < \frac{2300 - 2400}{500}\right) \\ &= P(Z < -0.2) \\ &= 0.4207 \end{aligned}$$

(c) (5pt) Use the answer in part (b) to compute the probability that at least four of five randomly selected high performance workstations of the same brand from dealers nationwide will cost less than \$2,300.

$$\begin{aligned} Y &= \# \text{ in } 5 \text{ that cost less than } \$2300 \\ &\sim \text{Bin}(5, \pi) \text{ where } \pi = 0.4207 \end{aligned}$$

$$\begin{aligned} P(Y \geq 4) &= P(Y = 4) + P(Y = 5) \\ &= \frac{5!}{4!1!} (0.4207)^4 (0.5793)^1 + \frac{5!}{5!0!} (0.4207)^5 (0.5793)^0 \\ &= 0.090733 + 0.013178 = 0.1039 \end{aligned}$$

(d) (5pt) The manufacturer of this workstation suspects that sales of their product is not competitive with similar machines in some areas because dealers have marked the price above the 90th percentile cut-off of the price distribution. What is this cut-off price?

$$\begin{aligned} c &= ? \text{ where } P(X > c) = 0.1 \Rightarrow P\left(Z > \frac{c - 2400}{500}\right) = 0.1 \\ \frac{c - 2400}{500} &= 1.28 \\ c &= 3040 \text{ i.e. } \$3040 \end{aligned}$$

(e) (5pt) Compute the probability that the average price of 100 high performance workstations randomly selected from dealers nationwide will cost between \$2,300 and \$2,450.

$$\bar{X} \sim N\left(2400, \frac{500^2}{100}\right)$$

$$\begin{aligned} P(2300 < \bar{X} < 2450) &= P\left(\frac{2300 - 2400}{500/\sqrt{100}} < Z < \frac{2450 - 2400}{500/\sqrt{100}}\right) \\ &= P(-2 < Z < 1) \\ &= 0.4772 + 0.3413 = 0.8185 \end{aligned}$$

4. Suppose that the observed readings of precision thermometers of a particular brand when the actual temperature measured is 65F, has mean μ and variance σ^2 . For the next five parts assume that a value of $\sigma^2=5.29$ is available from past data.

- (a) (4pt) Describe the sampling distribution of the sample mean random variable $\bar{Y}$ for a sample of size $n=36$, assuming that the distribution of readings from the population of thermometers is unknown. Justify your answer.

$$Y \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ where } \frac{\sigma^2}{n} = \frac{5.29}{36}$$

By CLT as n is large

- (b) (5pt) If you desire to estimate the population mean μ to be within 1F of the true value with 95% confidence, how many thermometers should you sample from the above population?

$$E=1 \quad z_{0.025} = 1.96 \quad \sigma^2 = 5.29$$

$$n = \frac{(z_{0.025})^2 \cdot \sigma^2}{E^2} = \frac{(1.96)^2 (5.29)}{1} = 20.32 \approx 21$$

The company statistician obtains a sample of 36 thermometers, and tests each to check for accuracy at 65F. The sample mean of the readings of the 36 thermometers is $\bar{Y}=63.8$.

- (c) (5pt) Construct a 95% confidence interval estimate of μ . Explain your result!

$$\bar{y} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$63.8 \pm 1.96 \cdot \frac{\sqrt{5.29}}{\sqrt{36}}$$

$$63.8 \pm 1.96 \cdot \frac{2.3}{6}$$

$$CI(63.05, 64.55)$$

We are 95% confident that the true population mean of temperature is between 63.05F and 64.55F.

- (d) (2pt) The management of the company wants you to construct a 99% confidence interval in order to obtain a more accurate estimate. Is this a reasonable proposal? Explain to the management why you agree or disagree.

The 99% CI will be wider than the 95% CI thus making the interval estimate less accurate.

- (e) (6pt) Test $H_0: \mu=65$ vs $H_a: \mu \neq 65$ using $\alpha=.01$. State the steps of the hypothesis test, the rejection region and your decision clearly.

$$z_c = \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}} = \frac{63.8 - 65}{2.3/\sqrt{6}} = -3.13$$

$$z_{0.01/2} = z_{0.005} = 2.575$$

Rejection region: $|z_c| > 2.575$

Since $|z_c| = 3.13$ is in the rejection region, we reject H_0 at $\alpha = 0.01$.

5. Suppose that 8% of the diodes used in motion detectors in a large case to be shipped from a factory are defective.

- (a) (1pt) Define the Binomial random variable X .

$X = \#$ of defective diodes in a case of 10
 $X \sim \text{Binom}(n, 0.08)$

- (b) (5pt) Compute the probability that at most 1 out of 10 diodes randomly selected from this case are defective.

$$\begin{aligned} P(X \leq 1) &= P(X=0) + P(X=1) \\ &= \frac{10!}{0!10!} (0.08)^0 (0.92)^{10} + \frac{10!}{1!9!} (0.08)^1 (0.92)^9 \\ &= (0.92)^{10} + 10 \cdot (0.08)(0.92)^9 = 0.4344 + 0.3778 \\ &= 0.8121 \end{aligned}$$

- (c) (6pt) An entire case will be rejected if more than 10 out of a random sample of 100 diodes tested from the case are defective. Use the Normal approximation to the Binomial to find the probability of rejecting a case if a random sample of 100 diodes from a case are tested.

$$n=100 \quad \pi=0.08 \quad n\pi=8 \quad n(1-\pi)=92$$

Need $P(Y > 10) = P(Y \geq 11)$ where $Y = \#$ defective diodes in 100

$$Y \sim N(n\pi, n\pi(1-\pi)) \quad Y \sim N(8, 7.36)$$

$$\begin{aligned} \text{So, } P(Y \geq 11) &\approx P\left(z > \frac{10.5 - 8}{\sqrt{7.36}}\right) = P(z > 0.92) \\ &= 1 - 0.8212 = 0.1788 \end{aligned}$$

6. If a new process for mining copper is to be put into full operation, it must produce an average of more than 50 tons of ore per day. A 15-day trial period was carried out and daily yields were recorded in tons per day. A sample mean of $\bar{Y} = 74.2$ and a sample standard deviation $s = 44.2$ were calculated from these data. Assume that the distribution of the yields per day is nearly Normal with mean μ .

- (a) (6pt) Write down appropriate null and alternative hypotheses needed in terms of μ to test whether on the average, the mine produces more than 50 tons of ore per day under the new process. Calculate a test statistic to test this hypothesis, state the rejection region, and your conclusion using $\alpha = .05$.

$$H_0: \mu \leq 50 \quad H_a: \mu > 50$$

$$t_c = \frac{74.2 - 50}{44.2 / \sqrt{15}} = 2.12$$

$$t_{0.05, 14} = 1.761 \Rightarrow \text{Rejection region: } t > 1.761$$

Since t_c is in the rejection region, we reject H_0 at the $\alpha = 0.05$ level, and conclude that the mine produces more than 50 tons of ore per day under the new process.

- (b) (3pt) Explain in words AND define by a formula what is meant by the p-value for the test in part (a) above.

$$p = P_{H_0}(T_{14} > 2.12) \text{ or}$$

probability of obtaining a t-value as large as 2.12 if H_0 is true.

- (c) (2pt) Approximately, compute (or bound) the p-value corresponding to part (b) using an appropriate table. Show your work!

From table under $df = 14$

$$0.025 < p < 0.05$$

- (d) (3pt) Show how to use the p-value you computed to test the hypothesis you stated in part (a).

Since $p < 0.05$, we reject H_0 at the $\alpha = 0.05$ level.

- (e) (7pt) Calculate a 90% confidence interval for μ . Use this interval to test the hypothesis you gave in part (a) (i.e., say if you would reject or fail to reject H_0 using the confidence interval). Explain how you arrived at your decision. What is the α -level of your test?

$$90\% \text{ CI: } 74.2 \pm t_{0.05, 14} \cdot \frac{44.2}{\sqrt{15}}$$

$$74.2 \pm 1.761 \cdot 11.4124$$

$$74.2 \pm 20.1$$

$$\text{CI}(54.1, 94.3)$$

Since 50 is not in this interval we reject H_0 at $\alpha = 0.05$

- (f) (3pt) Suppose the distribution of the yield per day is not normal, but their means and variance remain unchanged, will you still have the same results in part (a) and part (e)?

Why or why not? You don't need to do any calculations here!

Not in general, because when it is not normal we need CLT to justify the use of the normal distribution (approximation), which requires large sample size (at least 30). But in this case $n=15$ which is small.

7. Papa Husker's is a local pizza joint in Lincoln featuring fast delivery. Its number of delivery orders from 1:30pm-2pm on regular Saturdays roughly follows a Poisson distribution with $\lambda=7$.

- (a) (2pt) What are the expected value and variance for this distribution?

$$E(X) = 7$$

$$V(X) = 7$$

- (b) (4pt) Find the probability of having exactly 3 delivery orders in this time period on a regular Saturday.

$$P(X=3) = \frac{\lambda^3 e^{-\lambda}}{3!} = \frac{7^3 e^{-7}}{3!} = 0.052$$

- (c) (6pt) On a home game Saturday, this pizza joint received 15 delivery orders in this time period. Using normal approximation with continuity correction, what is the probability that this pizza joint receives the same number of or more orders between 1:30pm to 2pm on a regular Saturday? Write a conclusion based on your results.

$$\mu = 7 \quad \sigma^2 = 7$$

$$P(X \geq 15) = P(X \geq 14.5)$$

$$= P\left(\frac{X - \mu}{\sigma} \geq \frac{14.5 - 7}{\sqrt{7}}\right)$$

$$= P(Z \geq 2.83)$$

$$= 1 - P(Z < 2.83) = 1 - 0.9977 = 0.0023$$

The pizza joint is very unlikely to receive 15 or more delivery orders between 1:30pm and 2pm on a regular Saturday afternoon.